

PHYSICAL AND MORPHOLOGICAL CHARACTERIZATION OF UNDISTURBED AND DISTURBED PLOUGHSPANS IN A SANDY LOAM SOIL

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ABSTRACT

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In a sandy loam soil the ploughspan had a lower porosity and a higher bulk density and mechanical resistance than ploughspans that had been disturbed by rota-digging two years previously. Hydraulic conductivities (K) for pressure heads (h) from 0 to -20 cm were higher and moisture contents within the same h range were lower. Both values were identical for pressure heads lower than -20 cm. According to the widely used physical capillary pore model, the volume of pores $> 140 \mu\text{m}$ was 2% (v/v) in the undisturbed ploughspan and 8% (v/v) in the disturbed ploughspan. However, micromorphometric analyses showed that the undisturbed ploughspan contained the highest volume of large pores and many continuous root channels, while the disturbed ploughspan had no continuous large pores. Pore continuity was determined by using a staining test with methylene blue. Disturbance of the ploughspan in these soils results in formation of unstable fragments, which collapse, thereby disrupting the vertical continuity of the initially large packing voids between fragments.

INTRODUCTION

Ploughspans are common in medium textured cultivated soils in The Netherlands. They markedly affect soil moisture regimes and rooting patterns of agricultural crops and, thereby, agricultural productivity. Attempts have been made to remove ploughspans by various methods. In this paper the effects of deep tillage with a rota-digger are studied, comparing physical and micromorphological properties of a disturbed ploughspan two years after deep tillage with properties of the undisturbed ploughspan.

An important physical characteristic is the hydraulic conductivity (K), which determines drainage rates. Moisture content in a given ploughspan at a particular time and location is, of course, governed not only by K but also by moisture retention data. The latter are routinely obtained in many

laboratories. Measurement of K is, however, often more cumbersome, particularly when made for unsaturated soil. This study presents results obtained by applying the crust test for measuring K . This test is simple and requires little time. Thus, it becomes feasible to make multiple measurements within a reasonable time.

Thin sections of the different undisturbed soil materials were used for a quantitative analysis of pore size distributions by means of an image analyser, the Quantimet 720. Thin sections of samples which had been percolated with methylene blue were made and thus water-conducting pores could be recognized by their blue walls in the thin section.

In summary, the purpose of this study is to demonstrate: (1) differences in physical conditions between undisturbed and disturbed ploughpans in a sandy loam soil, (2) use of the crust test for measuring K and (3) application of micromorphological techniques to characterize porosity and flow patterns.

MATERIALS AND METHODS

Soils and land use

The soils being studied are located in the south-west of The Netherlands near Nieuw-Vossemeer and are developed in young marine deposits. They are classified as Typic Haplaquents (Soil Survey Staff, 1975) and as Calcaric Fluvisols (FAO—UNESCO, 1974). These soils have AC profiles and are well drained. They are under cultivation and have a rotation of potatoes, sugar beet and cereals, mainly winter wheat. In many of these cultivated soils compacted layers (ploughpans) have developed just below the tilled topsoil.

A short description of the studied pedon is given below and some analytical data are presented in Table I. The A_p horizon (0–31 cm) is dark grayish brown (2.5Y 3.5/2, M), has a sandy loam texture and a weakly developed structure, consisting of 5–20 mm wide, microporous, subangular blocky elements. The C_1 horizon (31–42 cm) is olive gray (5Y 5/2, M), has a sandy

TABLE I

Analytical data for a Typic Haplaquent with undisturbed ploughpan

Horizon	Depth (cm)	Sample depth (cm)	Particle size distribution (% w/w)						$CaCO_3$ (% w/w)	pH-KCl	Organic matter (% w/w)
			>150	105–150	50–105	16–50	2–16	<2 μm			
A_p	0–31	0–25	1.6	11.8	41.8	22.1	6.7	16.0	4.7	7.6	1.4
C_1	31–42	38–42	0.6	10.8	50.4	20.4	5.4	12.4	9.2	7.8	0.8
C_2	42–70	58–62	1.8	33.3	47.6	8.9	1.9	6.5	7.6	8.0	0.6

loam texture in which a massive, compact, ploughpan has developed. The C₂ horizon (42–70 cm) is grayish brown (2.5Y 5/2, M), with reddish mottles, it has a loamy sand texture and a very weakly developed structure consisting of 10–20 mm wide, macroporous, subangular blocky peds. Below 70 cm depth the soil consists of a laminated sand with reddish mottles. All horizons are calcareous.

Deep loosening by rota-digging in part of the field had created a mixture of A and C material within a depth of 50 cm. Between 0 and 30 cm depth about 80% of the soil material consisted of A and 20% of C material. Between 30 and 50 cm depth the percentages are 20% and 80%, respectively. A weakly to moderately developed structure of 5–20 mm wide, microporous, subangular blocky elements occurs in the loosened zone up to 50 cm depth.

The field was tile-drained; groundwater levels fluctuated between 0.80 m below the surface in the winter period and 2.0 m below the surface in the summer period.

Physical methods

Penetration resistances were measured in the field with a penetrometer with a registering device, producing instant output of the penetration resistance as a function of depth. The cone of the penetrometer had a surface area of 1 cm² and a top angle of 60°. The instrument is pushed down manually with a constant speed of approximately 2 cm s⁻¹.

Soil physical determinations included measurement of bulk density and porosity. Soil cores with a diameter of 9 cm and a height of 2 cm (150 cm³) or 4 cm (300 cm³) were sampled from an undisturbed soil column of 1 m length and 9 cm diameter, which was collected with a stainless steel cylinder pushed into the soil with a hydraulic pressure device. Special attention was paid to the measurement of moisture retention and hydraulic conductivity curves; these measurements will therefore be discussed in more detail.

The routine procedure for measuring moisture - retention data uses desorption of small soil-filled cylinders on a sand-bed. However, repeated weighing of samples may disrupt the contact between the soil in the cylinder and the sand-bed. As a result, hydraulic equilibrium conditions are sometimes not reached within the fixed time period between measurements. A low hydraulic conductivity of a soil sample may be another reason why equilibrium is not reached, even though the contact between soil and sand-bed may be perfect. To overcome these problems an alternative procedure can be followed (e.g., Baker et al., 1974). This procedure uses a closed porous cup which allows measurement of outflow from a single sample as a function of time. Starting at equilibrium for saturated conditions, outflow during successive pressure-drop increments is measured by observing water levels in a burette (Fig. 1).

Of particular interest is the hydraulic conductivity (*K*) of soil at and near saturation, which corresponds to field conditions in periods of the year

when precipitation exceeds evaporation. Several methods are available to measure K near saturation in soil with large pores (e.g., Bouma, 1983). The crust test is particularly suitable, because a large sample can be used, little error is involved and costs in terms of time and equipment are limited. Earlier versions of the crust test (Bouma et al., 1971; Bouma and Denning, 1974) have presented some problems due to unstable crust materials.

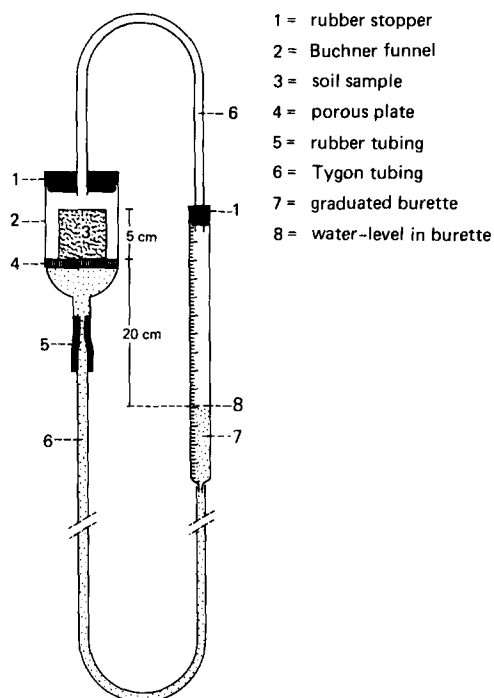


Fig. 1. Apparatus for measuring moisture-retention curves (after Baker et al., 1974).

The following procedures were used when applying the crust test in this study (see also Bouma et al., 1983): (1) Large, undisturbed samples of initially moist soil are collected in the field in plastic cylinders with a diameter of 20 cm. The cylinder has a sharpened lower edge, and, in our case, a height of 20 cm. It is gently pushed into a column of soil which has been prepared in situ. Thus, disturbances are minimized. (2) In the laboratory, the soil-filled cylinders are placed on a screen, which, in turn, rests on a funnel, allowing collection of outflow. Two tensiometers are placed below the upper soil surface at 2 and 4 cm depth, respectively, to measure sub-crust pressure heads. (3) Crusts are made, using mixtures of coarse sand and quick-setting cement which can be bought in any hardware store. Sand and cement are thoroughly mixed when dry, water is added and mixing is continued until a thick paste is formed which is evenly applied to the upper soil surface, ensuring perfect contact. Thus, stable crusts are formed which do

not dissolve. Crusts with a low hydraulic resistance contain 5–10% (v/v) of cement; higher resistances correspond to cement contents of 20% (v/v) and above. A series of crusts is applied.

A run is made for each crust (starting with the lightest) and consists of the measurement of a steady flux in terms of an outflow rate, a stable subcrust pressure head (h) and a pressure head gradient. The same cylinder is used with different crusts. The flux is equal to K_{unsat} at grad $H = 1 \text{ cm cm}^{-1}$, at a subcrust pressure head of $-h \text{ cm}$ (H is the hydraulic head). In our experiments grad H was 1 cm cm^{-1} and shallow ponding on top of the crust was maintained. Measurement of outflow rates is simpler than measuring inflow rates with an infiltrometer. Also, laboratory measurements in series are faster than separate field measurements. Of course, K_{unsat} values cannot be obtained at pressure heads lower than those corresponding to the height of the column, which can be varied as needed. It may, however, be more practical to place the 20 cm cores on a container filled with sand, thereby extending the depth of the flow system. Then, inflow rates must be measured, which is more complicated.

The crust test is particularly suitable to measure K_{unsat} near saturation in soils with macropores, where a sharp drop of K occurs due to emptying of macropores. K at zero pressure is referred to as K_{sat} when macropores are filled with water and as $K_{(\text{sat})}$ when macropores are filled with air (Bouma, 1982). K_{sat} can be obtained by measuring the steady flux through uncrusted soil.

Sampling and staining

Samples for micromorphological analysis were collected in stainless steel boxes ($16 \times 15 \times 8 \text{ cm}$). A soil column of slightly larger dimensions was carefully carved out in situ and the boxes were gently pushed into the column in the same way as the samples were collected for the crust test. Two sets of boxes, one containing the undisturbed and the other the disturbed ploughpan, were treated in the same way at the same time. Samples were taken at the same depth and at adjacent locations as those for K measurements.

The soil in the boxes was slowly saturated in the laboratory from the bottom upwards to avoid entrapment of air. K_{sat} was determined by ponding water on top and by measuring the flux. This measurement was made to see whether these values compared well with K measurements made in the large cylinders. They usually did, and if so, a solution of 10% methylene blue was ponded on top of the sample. Ponding was continued until the colour of the influent and effluent had the same intensity (Bouma et al., 1977; 1979). Next, the columns were cut into five horizontal slices of 2 cm thickness using a thin metal wire, and thin sections were prepared.

Micromorphological methods

Soil structure and porosity can be studied in detail in thin sections. Thin sections are made of blocks of undisturbed soil material, in this case $15 \times 8 \times 2$ cm, which are air-dried or oven-dried at 40°C and impregnated with a plastic. After hardening, these blocks are cut and slabs from the central parts are ground to $20\ \mu\text{m}$ thick slices and mounted on glass slides (Jongerius and Heintzberger, 1975).

Photograms, special photographs in which only voids are black, can be made of the entire thin section (Ismail, 1975). These photograms, which were made with the film POL type 55 (50 ASA) with $2 \times$ magnification, are used for morphometric porosity analysis with an electro-optical image analyser, the Quantimet 720 (Ismail, 1975). Several soil-pore parameters, e.g., area, perimeter, intercept and number, can be measured. Using the pattern recognition system, individual pores can also be characterized. More detailed information on the application of the Quantimet 720 in soil studies is presented by Ismail (1975), Jongerius (1974, 1975) and Bouma et al., (1979). Each photogram was measured in 24 fields and results compared to check the homogeneity, to ensure that representative surfaces were studied. Pore diameters were calculated from area measurements. Six classes are distinguished (see Fig. 5). Pores with blue-stained walls were measured separately, using identical procedures.

RESULTS AND DISCUSSION

The undisturbed ploughpan had a higher penetration resistance than the disturbed ploughpan (Fig. 2). Bulk densities were highest in the upper part of the compacted layer and porosities were lowest (Table II).

Moisture-retention curves, obtained from seven locations for each treatment, are shown in Fig. 3. The decrease of the moisture content from saturation ($h = 0$ cm) to $h = -20$ cm was largest in the disturbed ploughpan (B). On average, moisture content at $h = -20$ cm and the decrease of the moisture content from $h = -20$ cm to $h = -100$ cm were not significantly different. The loss of water from $h = 0$ cm to $h = -20$ cm is a measure of the volume of soil pores in the (equivalent) pore size classes $> 140\ \mu\text{m}$, according to the capillary pore model. The moisture-retention curves thus suggest the occurrence of 8% (v/v) of pores $> 140\ \mu\text{m}$ in the disturbed ploughpan. Conversely, pores of similar size would occupy only 2% (v/v) in the undisturbed ploughpan.

Hydraulic conductivity curves, obtained with the crust test, are shown in Fig. 4. On average, K_{sat} was significantly higher in undisturbed ploughpans and the drop from K_{sat} to K at $h = -5$ cm was larger. The latter phenomenon is due to emptying of continuous large pores, such as root channels, which may induce a relatively high K_{sat} when filled with water. However, these large pores become air-filled at negative pressure heads very close to satura-

tion (e.g. Bouma, 1982). K values for the undisturbed ploughpan were significantly higher than those for the disturbed ploughpan, but differences for pressure heads lower than -10 cm were minor. The observed differences between K_{sat} values have practical significance because at K_{sat} values < 5 cm

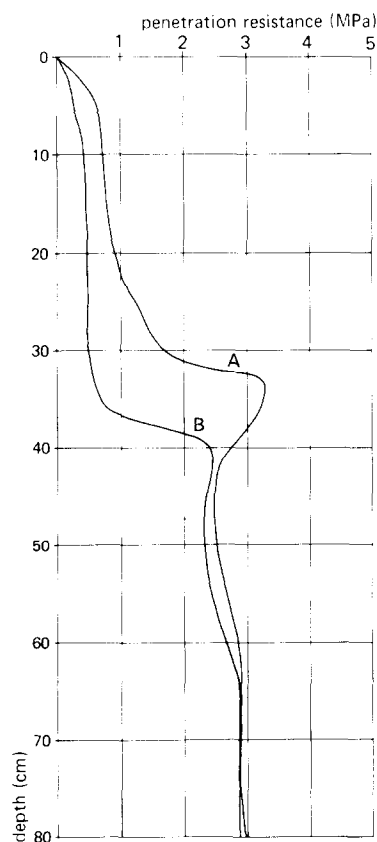


Fig. 2. Mechanical resistances as measured with a penetrometer in the soil with the undisturbed ploughpan (A) and in the soil with the disturbed ploughpan (B), November 1982.

TABLE II

Bulk density and porosity of soil with an undisturbed ploughpan (A) and of soil with a disturbed ploughpan (B)

Depth (cm)	Bulk density (g cm^{-3})		Porosity ($\text{cm}^3 \text{cm}^{-3}$)	
	A	B	A	B
31–33	1.70	1.58	0.36	0.40
33–35	1.63	1.59	0.39	0.40
35–39	1.58	1.58	0.40	0.40
40–44	1.55	1.58	0.42	0.40

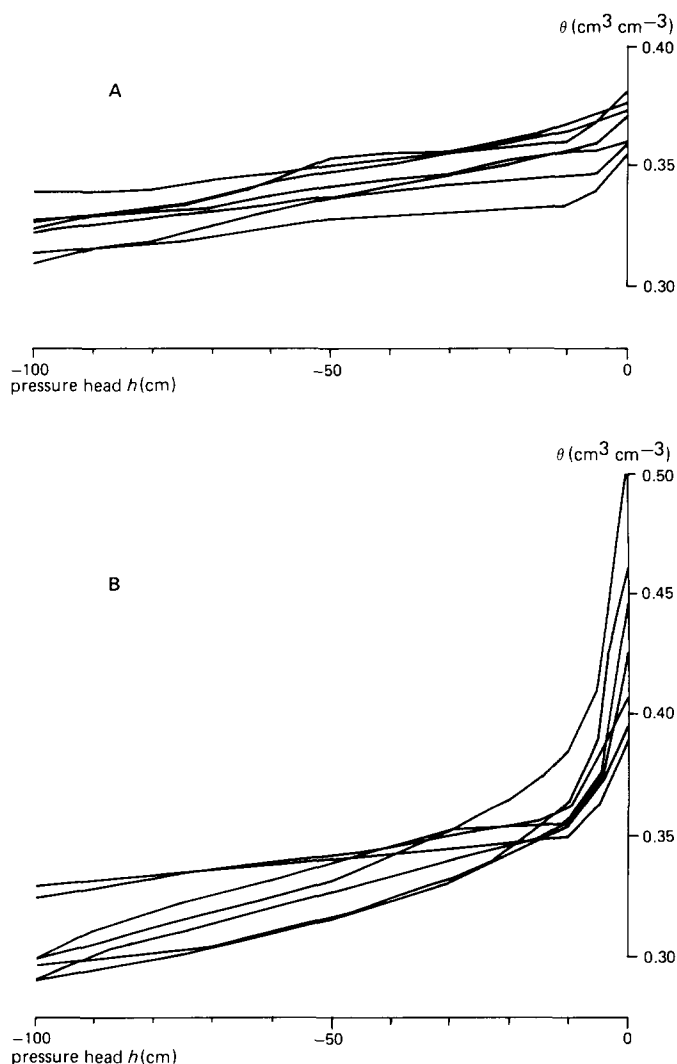


Fig. 3. Moisture—retention curves derived from seven samples from soil with the undisturbed ploughpan (A) and of the soil with the disturbed ploughpan (B).

per day, ponding and perched water tables may occur. This assumption is based on an average measured shower intensity in winter of 2 mm h^{-1} . Thus, the lower K_{sat} values of the disturbed ploughpan indicate a higher probability for occurrence of perched water tables.

Results of the image analyses are presented in Fig. 5. Two years after disturbance of the ploughpan, the total volume of pores $> 30 \mu\text{m}$ was approximately 3% (v/v), which was lower than the corresponding volume of approximately 6% (v/v) in the undisturbed ploughpan. Volumes of pores between 30 and $300 \mu\text{m}$ were similar. The difference between both treatments was therefore mainly due to pores $> 300 \mu\text{m}$, which occur in a sig-

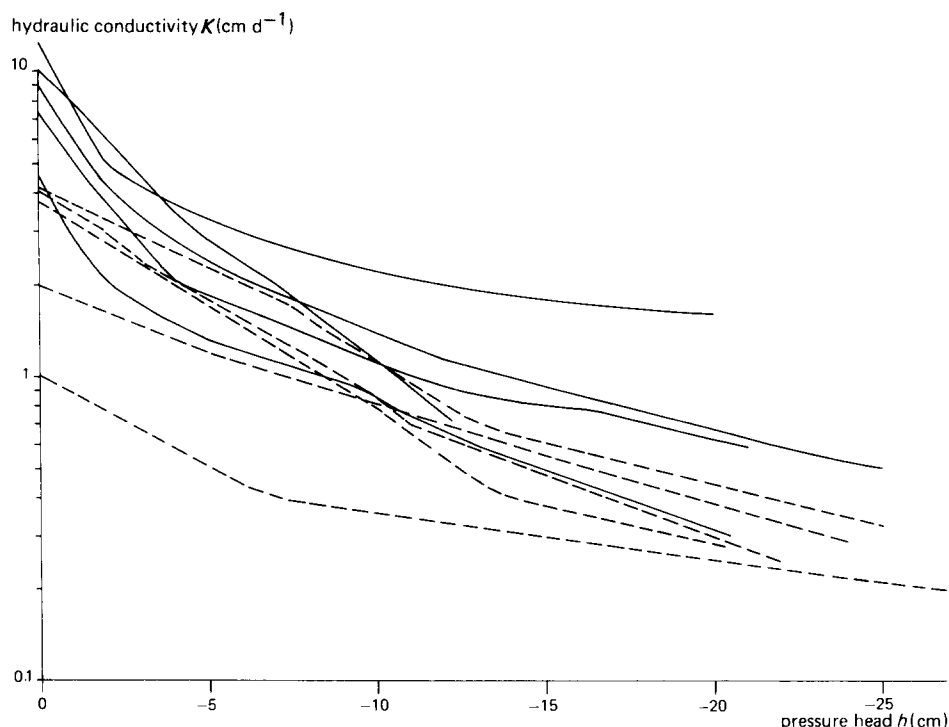


Fig. 4. Hydraulic conductivity curves derived from five samples from soil with the undisturbed ploughpan (—) and with the disturbed ploughpan (- - -).

nificantly higher quantity in the undisturbed ploughpan. Blue-stained pores, indicating their activity in conducting water, occurred at all levels in the undisturbed ploughpan, occupying approximately 1–2% (v/v). Only large pores with diameters $> 500 \mu\text{m}$ had stained walls. Except for the upper sample, no stained pores were found in the disturbed ploughpan.

Two representative pictures, obtained from the thin sections at 39 cm depth, illustrate the differences observed (Fig. 6). In the undisturbed ploughpan (A) large stained pores occur in a relatively homogeneous groundmass. The disturbed ploughpan is characterized by a heterogeneous groundmass, consisting of a mixture of A and C material, with some larger but unstained pores. The pores present in the undisturbed ploughpan are mainly vertically oriented root channels. In the disturbed ploughpan most pores consist of voids resulting from the setting of the soil after rota-digging. These are isolated voids, which do not form a continuous pore system.

Pore volumes were highest in the disturbed ploughpan, but volumes of observed large pores were lowest. The difference must therefore be due to a larger volume of pores $< 30 \mu\text{m}$ diameter occurring in the disturbed groundmass, possibly as a result of puddling during deep tillage. More water was extracted in the pressure range from $h = 0 \text{ cm}$ to $h = -20 \text{ cm}$.

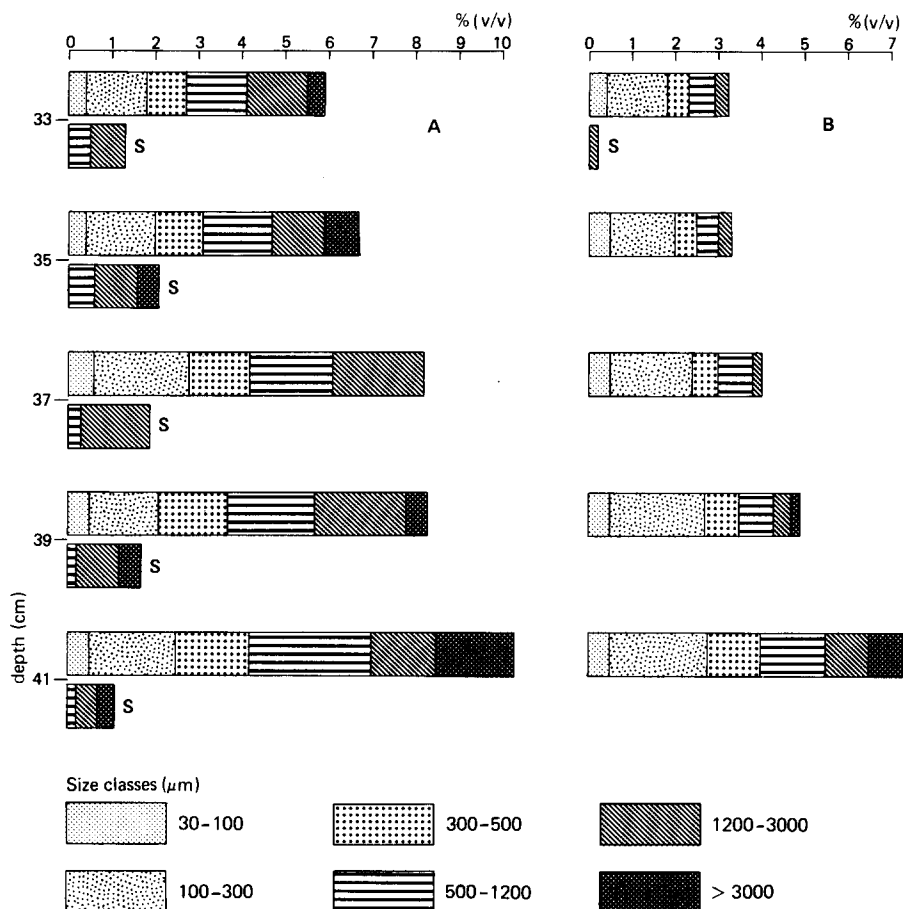


Fig. 5. Porosity measurements performed with the Quantimet 720 on horizontal thin sections of stained soil columns in the zone containing the undisturbed ploughpan (A) and the corresponding depth in the soil with the disturbed ploughpan (B). The upper bar for each depth shows the total porosity of pores larger than $30\ \mu\text{m}$ diameter, the lower bar (S) shows only the pores with blue-stained walls.

Obviously, however, this does not indicate occurrence of large pores as might be inferred from the capillary-pore model. The latter model should therefore be used with care when predicting hydraulic flow parameters.

CONCLUSIONS

(1) Rota-digging of a ploughpan in a sandy loam soil did not result in improved hydraulic properties, as observed after two years. Hydraulic conductivity for pressure heads down to $-20\ \text{cm}$ was significantly lower as compared with the undisturbed ploughpan.

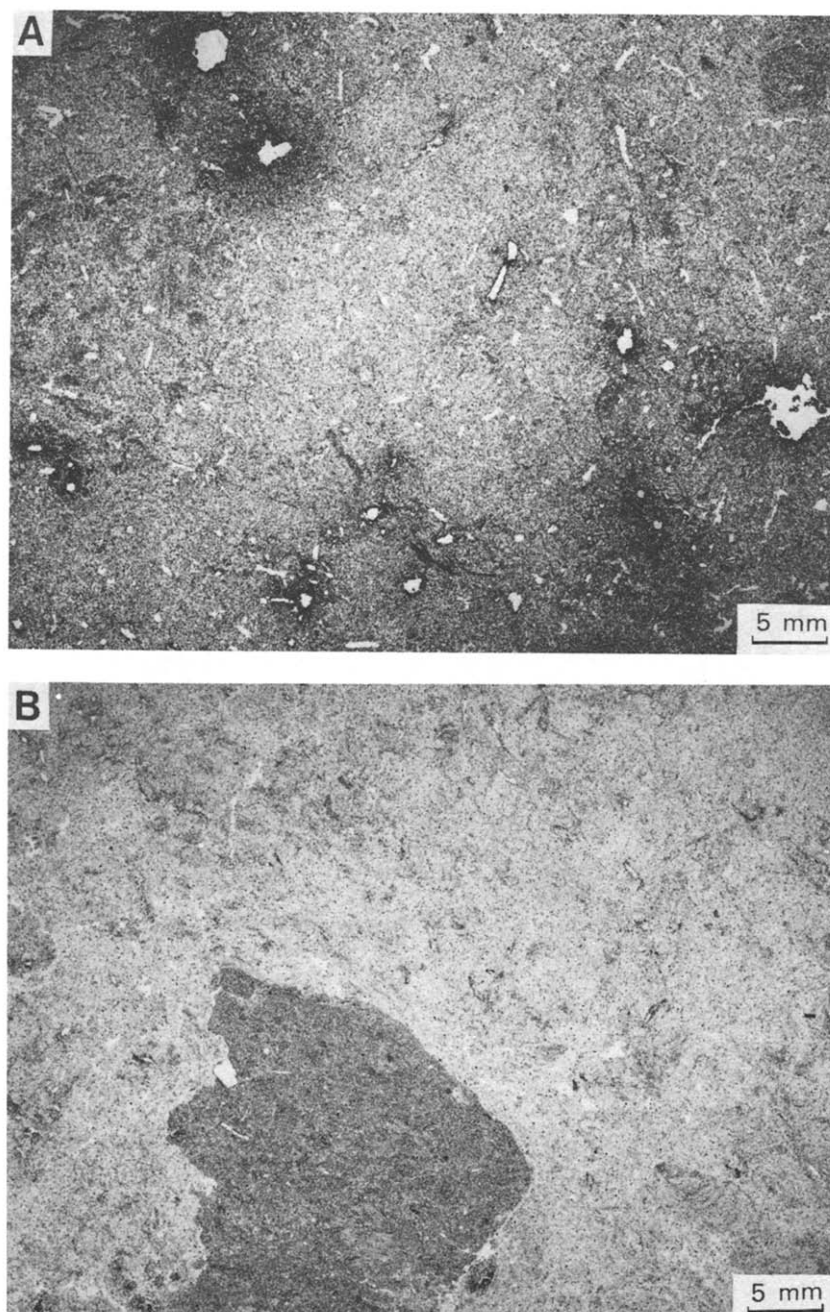


Fig. 6. Photographs of horizontal thin sections of the stained soil columns at 39 cm depth of the undisturbed ploughpan (A) and the corresponding depth in the soil with the disturbed ploughpan (B) (porosity measurements are shown in Fig. 5). Stained pores (S) occur only in the ploughpan.

(2) Rota-digging resulted in a lower bulk density and a higher porosity. This loosening effect was also observed in terms of a decreased penetrometer resistance. Accordingly, moisture-retention curves for the disturbed ploughpan showed a stronger decrease of moisture contents from saturation down to a pressure head of -20 cm, as compared with the undisturbed ploughpan. This, in turn, would mean a higher volume of pores $> 140 \mu\text{m}$, according to the standard equivalent pore-size model. However, this observation is at odds with the measured lower K values and the morphometric analyses.

(3) Micromorphometric data showed that the disturbed ploughpan contained fewer large pores with a diameter $> 300 \mu\text{m}$. More importantly, no stained pore walls were observed in the methylene-blue experiments, in contrast to the undisturbed ploughpan, which had stained large pores with a diameter $> 500 \mu\text{m}$. Lack of stained pores indicates a low vertical large-pore continuity, which explains the measured lower K . The physical equivalent pore-size model should be used with care when predicting hydraulic flow parameters, which, as demonstrated in this study, are strongly influenced by large-pore continuity.

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